

CSS3 Exercise Answers

Local Community Event Portal

Module 1 — Complete Solutions with Code Examples

[CSS3 Exercises — All 11 Topics Covered](#)

EXERCISE 1

Why CSS? Inline vs. Internal vs. External

Scenario: The designer wants you to experiment with different ways to apply styles.

① Inline Style — Red Heading

Add the **style** attribute directly on the HTML tag. Use only for one-off overrides.

```
<h1 style="color: red;">Community Event Portal</h1>
```

② Internal (Embedded) Style — Body Background

Place a **<style>** block inside **<head>**. Useful for single-page styling.

```
<head>
<style>
/* Internal style for body background */
body {
background-color: #f0f4f8;
margin: 0;
font-family: Arial, sans-serif;
}
</style>
</head>
```

③ External Stylesheet — styles.css

Link an external file from **<head>**. Best practice for reusable, maintainable styles.

```
<!-- In index.html -->
<head>
<link rel="stylesheet" href="styles.css">
</head>

/* styles.css */
/* Header styles */
h1 {
font-size: 2rem;
color: #0f3460;
}

/* Navigation styles */
nav a {
text-decoration: none;
color: #e94560;
}

/* Card styles */
.eventCard {
border: 1px solid #ddd;
padding: 16px;
border-radius: 8px;
}
```

■ Prefer external CSS for projects with multiple pages — changes in one file apply everywhere.

CSS Syntax and Comments

Structured styles.css with Comments

■ *Rule of thumb: comment every major section. Use consistent 2- or 4-space indentation.*

EXERCISE 3

Selectors Playground

Scenario: Style elements based on IDs, classes, and element types.

All Selector Types Demonstrated

```
/* Universal selector – reset margin & padding */
* {
margin: 0;
padding: 0;
box-sizing: border-box;
}

/* Element selector – all h2 headings */
h2 {
font-size: 1.5rem;
color: #0f3460;
margin-bottom: 12px;
}

/* ID selector – unique banner */
#mainHeader {
background: linear-gradient(135deg, #0f3460, #e94560);
color: white;
padding: 40px;
text-align: center;
}

/* Class selector – event cards */
.eventCard {
background: #fff;
border: 1px solid #ddd;
border-radius: 8px;
padding: 20px;
margin-bottom: 16px;
box-shadow: 0 2px 6px rgba(0,0,0,0.08);
}

/* Grouping selector – h3 and p share spacing */
h3, p {
margin-bottom: 8px;
line-height: 1.5;
}
```

EXERCISE 4

Color & Background Styling

Scenario: Theme the portal based on a city council's branding.

```
/* HEX color for text */
body {
color: #1a1a2e; /* dark navy text */
}

/* RGBA – semi-transparent overlay */
.overlay {
background-color: rgba(15, 52, 96, 0.75); /* blue at 75% opacity */
color: white;
padding: 30px;
}

/* Background image with fallback color */
body {
background-color: #e8ecf0; /* fallback */
background-image: url('city-bg.jpg');
background-size: cover;
background-position: center;
background-attachment: fixed;
}

/* Linear gradient on section headers */
.sectionHeader {
background: linear-gradient(135deg, #0f3460 0%, #e94560 100%);
color: white;
padding: 24px 32px;
border-radius: 6px;
}

/* Diagonal gradient variant */
.heroSection {
background: linear-gradient(to bottom right, #1a1a2e, #0f3460);
min-height: 300px;
}
```

■ Always provide a solid background-color fallback before background-image in case the image fails to load.

EXERCISE 5

Typography: Fonts and Text

Scenario: The marketing team wants more appealing fonts and better readability.

```
/* Import Google Font in HTML <head> */
<link href="https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&family=Playfair+Display:wght@700&display=swap" rel="stylesheet">

/* OR via @import at top of CSS file */
@import url('https://fonts.googleapis.com/css2?family=Roboto&display=swap');

/* ■■ Font properties ■■ */
body {
  font-family: 'Roboto', Arial, sans-serif;
  font-size: 16px;
  font-weight: 400; /* normal */
  font-style: normal;
}

h1, h2 {
  font-family: 'Playfair Display', Georgia, serif;
  font-weight: 700;
  font-style: normal;
}

.tagline {
  font-style: italic;
  font-weight: 300;
  font-size: 1.1rem;
}

/* ■■ Text appearance ■■ */
.eventCard p {
  text-align: justify;
  line-height: 1.7;
  letter-spacing: 0.02em;
}

.sectionTitle {
  text-align: center;
  text-transform: uppercase;
  letter-spacing: 0.15em;
  font-size: 0.9rem;
  color: #e94560;
}

nav a {
  text-transform: capitalize;
  letter-spacing: 0.05em;
}
```

EXERCISE 6

Link and List Styling

Scenario: The default blue links and bullet lists don't match the design.

Link Pseudo-classes

```
/* Unvisited link */
a:link {
  color: #e94560;
  text-decoration: none;
}

/* Visited link */
a:visited {
  color: #9b2335;
}

/* Mouse hover */
a:hover {
  color: #0f3460;
  text-decoration: underline;
  transition: color 0.3s ease;
}

/* Click moment */
a:active {
  color: #ff0000;
}
```

List Styling

```
/* Remove bullets from nav */
nav ul {
  list-style-type: none;
  padding: 0;
  margin: 0;
  display: flex;
  gap: 20px;
}

nav ul li {
  padding: 8px 0;
}

/* Custom bullet for event lists */
.eventList {
  list-style-type: square;
  list-style-position: inside;
  padding-left: 16px;
}

.eventList li {
  padding: 6px 0;
  margin-bottom: 4px;
  border-bottom: 1px solid #eee;
}
```

EXERCISE 7

Table Styling

Scenario: The events admin table needs a cleaner look.

```
/* Reset and basic styling */
table {
width: 100%;
border-collapse: collapse; /* removes double borders */
font-size: 0.95rem;
margin-top: 20px;
}

/* Header row */
th {
background-color: #0f3460;
color: white;
padding: 12px 16px;
text-align: center;
letter-spacing: 0.05em;
text-transform: uppercase;
font-size: 0.85rem;
}

/* Data cells */
td {
padding: 10px 16px;
text-align: center;
border-bottom: 1px solid #e0e0e0;
}

/* Zebra striping – even rows */
tr:nth-child(even) {
background-color: #f2f6fc;
}

/* Hover highlight */
tr:hover {
background-color: #dce8ff;
transition: background-color 0.2s;
}

/* First column left-align */
td:first-child {
text-align: left;
font-weight: 600;
}
```

■ Always use `border-collapse: collapse` on the table, not on `tr`/`td` — it only works on the table element.

EXERCISE 8

Box Model & Layout Control

Scenario: Sections are cramped and need spacing.

Box Model for .eventCard

```
/* Box model properties */
.eventCard {
width: 320px;
padding: 20px 24px; /* inner spacing */
border: 2px solid #0f3460; /* visible border */
margin: 16px auto; /* outer spacing */
border-radius: 10px;
box-shadow: 0 4px 12px rgba(0,0,0,0.1);
}

/* Outline on focused form fields (doesn't affect layout!) */
input:focus,
textarea:focus {
outline: 3px solid #e94560;
outline-offset: 2px;
}
```

visibility: hidden vs display: none

```
/* visibility: hidden – element is INVISIBLE but still occupies space */
.hiddenBadge {
visibility: hidden;
}

/* display: none – element is REMOVED from layout (no space taken) */
.removedSection {
display: none;
}

/* Tip: toggle display with JS for show/hide menus */
.dropdown.active {
display: block;
}
```

■ Use `visibility:hidden` when you need the space preserved (e.g. a placeholder). Use `display:none` to truly hide and collapse the element.

EXERCISE 9

Multiple Columns in Text

Scenario: The community bulletin needs to be displayed like a newspaper.

HTML Structure

```
<section class="bulletin">
<h2>Community Bulletin</h2>
<article class="news-content">
<p>The city council has announced three new events
planned for the summer season...</p>
<p>Residents are encouraged to participate in the
annual cleanup drive scheduled for July...</p>
<p>Local sports teams are preparing for the inter-
ward championship starting next month...</p>
</article>
</section>
```

CSS Multi-Column Layout

```
/* Newspaper-style multi-column */
.news-content {
column-count: 2; /* number of columns */
column-gap: 30px; /* space between columns */
column-rule: 1px solid gray; /* dividing line */
text-align: justify;
font-size: 0.95rem;
line-height: 1.7;
}

/* Headline spans all columns */
.news-content h3 {
column-span: all;
text-align: center;
margin-bottom: 16px;
}

/* Responsive: single column on small screens */
@media (max-width: 600px) {
.news-content {
column-count: 1;
}
}
```

EXERCISE 10

Responsive Web Design with Media Queries

Scenario: Users will access the portal on phones, tablets, and desktops.

Base Layout (Desktop)

```
/* Desktop default — horizontal nav */
nav ul {
  display: flex;
  flex-direction: row;
  gap: 24px;
  list-style: none;
  padding: 0;
}

.container {
  width: 90%;
  max-width: 1200px;
  margin: 0 auto;
}

img {
  max-width: 100%;
  height: auto;
}

h1 { font-size: 2.5rem; }
p { font-size: 1rem; }
```

Media Query — Tablet (<768px)

```
@media (max-width: 768px) {
  /* Stack nav vertically */
  nav ul {
    flex-direction: column;
    align-items: center;
    gap: 10px;
  }

  h1 { font-size: 1.8rem; }
  p { font-size: 0.95rem; }

  .eventCard {
    width: 100%;
  }
}
```

Media Query — Mobile (<480px)

```
@media (max-width: 480px) {
  body {
    font-size: 14px;
  }

  h1 { font-size: 1.4rem; }

  img {
```

```
width: 100vw;
}

.container {
width: 95%;
padding: 0 10px;
}
}
```

Bonus: Flexbox & Grid

```
/* Flexbox card row */
.cardRow {
display: flex;
flex-wrap: wrap;
gap: 20px;
justify-content: center;
}

.eventCard {
flex: 1 1 280px; /* grow, shrink, min-width */
max-width: 360px;
}

/* CSS Grid layout */
.eventsGrid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
gap: 24px;
}
```

■ *Mobile-first approach: write base styles for mobile, then use min-width queries to scale up for larger screens.*

EXERCISE 11

Debug and Test with DevTools and VS Code

Scenario: Some users report layout breaking on smaller screens.

Step-by-Step DevTools Workflow

Open DevTools — Right-click any element → **Inspect** (or F12 / Ctrl+Shift+I).

Device Toolbar — Click the **Toggle Device Toolbar** icon (Ctrl+Shift+M). Select from preset devices (iPhone, iPad, Galaxy) or enter a custom width.

Inspect Styles — Click an element in the Elements panel. The **Styles** tab shows all applied CSS rules with their source file and line number. Crossed-out rules are overridden.

Live Edit — Click any CSS value in the Styles panel and type a new value — changes appear instantly. Great for testing colours, sizes, and spacing.

Computed Tab — Shows the **final calculated** value for every property after all cascades. Use the box model diagram to verify margin/padding.

Network Tab — Reload the page (Ctrl+R) while DevTools is open. Find **styles.css** in the list — a green 200 status means loaded; a red 404 means the path is wrong.

Console Tab — Run `document.querySelector('.eventCard').style` to inspect inline styles programmatically.

VS Code Tips for CSS

```
/* 1. Emmet shortcuts – type and press Tab:
bg → background;;
m10 → margin: 10px;
p20 → padding: 20px;

2. CSS variable hover: hover over var(--color) to see the value.

3. Auto-format: Shift+Alt+F

4. Live preview: Install the "Live Server" extension,
then right-click index.html → Open with Live Server.
The browser auto-refreshes on every save. */
```

Common Debug Checklist

- Is the `<link>` tag inside `<head>` with the correct href path?
- Check spelling of class/ID names — CSS is case-sensitive.
- Is a more specific selector overriding your rule? Check the Styles panel.
- Did you forget a semicolon at the end of a property value?
- Missing closing brace ``}`` can break all rules below it.
- Use **`border: 1px solid red`** as a temporary debug border to visualize element size.

Quick Reference — All Topics

#	Topic	Key Property / Concept
1	Inline / Internal / External	style attr • <style> • <link>
2	CSS Syntax & Comments	/* comment */ • selector { prop: val; }
3	Selectors	* • element • #id • .class • grouping
4	Colors & Backgrounds	HEX • RGBA • linear-gradient()
5	Typography	@import • font-family • text-transform
6	Links & Lists	:hover • :visited • list-style-type
7	Tables	border-collapse • nth-child(even)
8	Box Model	margin • padding • border • outline
9	Multi-Column	column-count • column-gap • column-rule
10	Responsive Design	@media • flexbox • grid • vw/vh/%
11	DevTools & Debug	Inspect • Device Toolbar • Network tab